



Installation Instruction

Installation Instruction Number: 5414654

Released Date: 05-nov-2018

Steel Fuel Supply Tube Installation Instructions

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Product Affected

- ISX15 CM2350 X101
- X15 CM2350 114B

The purpose of this document is to provide installation instructions for two newly released fuel supply tube kits; plumbing fuel from the low-pressure pump to the secondary fuel filter. New steel fuel supply tube(s) with “flexible” section have been introduced for increased robustness against demanding operating conditions. Improved fuel supply tubes can be retrofitted to existing nylon fuel supply tube configuration(s). See Figure 1 and Figure 2 for visual representations.

It is **not** necessary to remove the fuel filter for this installation.

Parts Required

Configuration A:

There may be an overlap in engine serial number (ESN). Verify the vehicle’s appropriate fuel supply tube configuration and part number is ordered before commencing repairs. See Figure 1 and Figure 2 for visual confirmation and Table 3 for part number verification.

Service procedure for approximate engine serial number (ESN) range: 79890940 to 80038419.

A service kit is available and includes components required to install steel fuel supply tube. See Table 1 below.

Table 1, Service Kit, Part Number 5579015

Part Description	Part Number	Quantity
Tube, Fuel Supply	5575157	1
Screw, Hex Flange Head Cap	3900631	1
Screw, Banjo Connector	3924725	2
Washer, Sealing	3963988	4
Clip	3685612	1

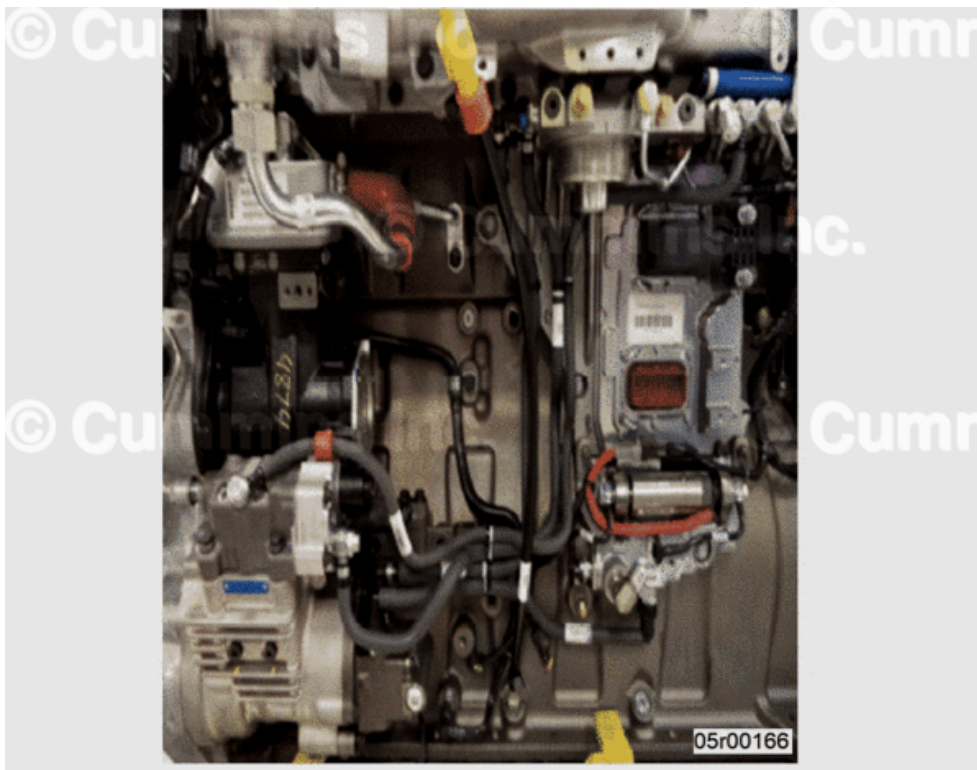


Figure 1, Representation of Configuration A.

Configuration B:

There may be an overlap in engine serial number (ESN). Verify the vehicle's appropriate fuel supply tube configuration and part number is ordered before commencing repairs. See Figure 1 and Figure 2 for visual confirmation and Table 3 for part number verification.

Service procedure for approximate engine serial number (ESN) range: 79999784 - 80088768.

A service kit is available, Part Number 5473412, includes components required to install steel fuel supply tube. See Table 2, Service Kit 5473412 structure.

Table 2, Service Kit, PArt Number 5473412

Part Description	Part Number	Qty.
Tube, Fuel Supply	5558764	1
Screw, Banjo	3924725	2
Washer, Sealing	3963988	4
Clamp, Tube	3975497	1
Clip	3685612	1



Figure 2, Representation of Configuration B

Steel fuel supply tube(s) have obsoleted & superseded previous fuel supply tube iteration(s). See Table 2.

Table 3, Service Parts

Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Tube, Fuel Supply Configuration A	3691498	Yes	Yes	5575157
Tube, Fuel Supply Configuration B	5440474	Yes	Yes	5558764

Personal Protection Equipment (PPE)

- Completing a Job Safety Assessment (JSA) prior to performing work helps identify job safety hazards and prevent incidents.
- To reduce the possibility of personal injury, personal protective equipment (PPE) must be utilized. Reference General Safety Instructions Procedure 204-006 in Section I of the corresponding Service Manual.

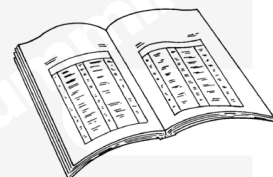
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

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⚠ WARNING ⚠

Do not vent the fuel system on a hot engine; this can cause fuel to spill onto a hot exhaust manifold, which can cause a fire.

- Chock vehicle wheels or use an appropriate anti-roll device to stabilize vehicle.
- Open vehicle hood.
- Disconnect the batteries. See Original Equipment Manufacturer (OEM) service information.
- Do **not** remove the fuel filter during installation.

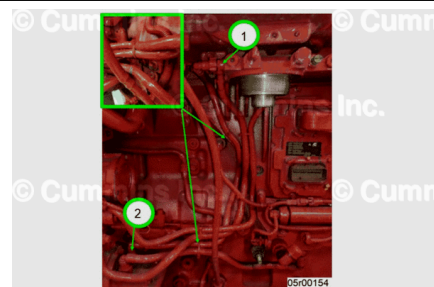
Configuration A

Remove

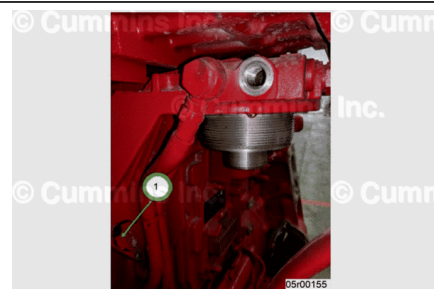
Place an appropriate drip pan or absorbent mat directly under the low-pressure fuel supply tube plumbing to contain any fuel that may discharge during the repair process.

Clean area around the fuel pump gear pump and fuel filter supply tube connections.

- Clip zip ties securing fuel supply tube part number 3691498; plumbing fuel from the low-pressure pump to the fuel filter inlet.
- Disconnect and discard fuel supply tube.
- Remove Quick Disconnect fittings associated with detached fuel supply tube.
 - At fuel filter inlet (1)
 - At fuel pump gear pump outlet (2)

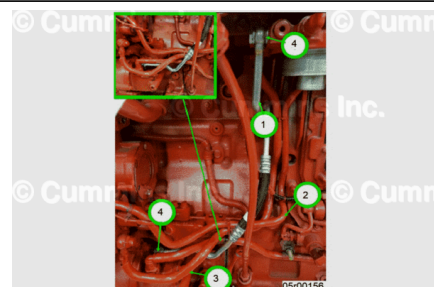


Remove and discard zip tie clip from fuel filter head bolt boss (1).



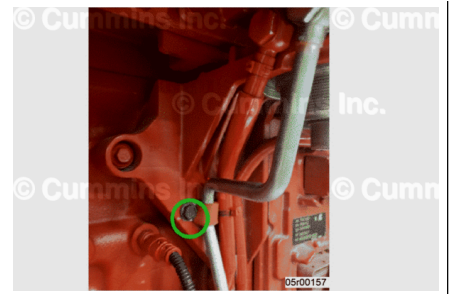
Install

- Route steel fuel supply tube (1) part number 5575157 under the fuel pump drain line (2) and filter to fuel pump (3) plumbing.
- Install banjo screw part number 3924725 with 2x sealing washer part number 3963988 hand tight, into fuel supply tube banjo(s):
 - At fuel filter inlet (4).
 - At fuel pump gear pump outlet (5).



- Install P-clip part number 3685612 onto steel fuel supply tube.
- Mount fuel supply tube and P-clip assembly onto fuel filter using 10mm bolt.

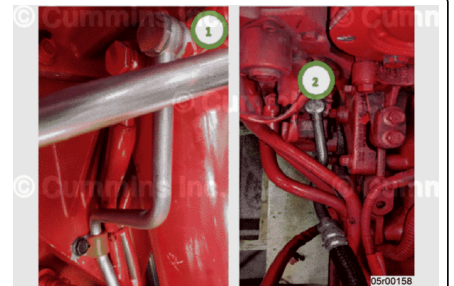
Torque Value: 23 n•m [17 ft-lb]



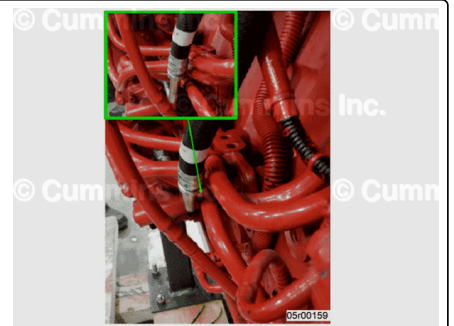
Torque banjo screws.

- At fuel filter inlet (1).
- At fuel pump gear pump outlet (2).

Torque Value: 34 n•m [25 ft-lb]

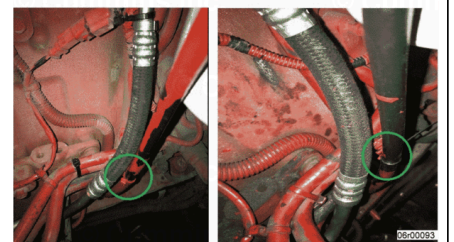


- Using a zip tie, secure fuel supply tube plumbing fuel from the electric lift pump to the fuel pump gear pump inlet using existing engine block clip. Ensure there is no contact between steel supply tube and neighboring nylon fuel supply tubes or engine oil dip stick tube.
- If engine oil dip stick tube is contacting the steel supply tube, clip and reposition looped zip tie adjusting the dipstick tube to provide adequate clearance.



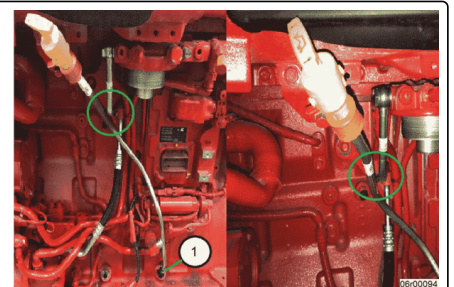
If engine oil **nylon** dip stick tube is contacting the steel supply tube, to provide adequate clearance:

- Cut and reposition looped zip tie along nylon fuel supply tube.



If engine oil **steel** dip stick tube is contacting the steel supply tube, to provide adequate clearance:

- Loosen dipstick tube base nut (1) adjusting its orientation to achieve adequate clearance.
- Tighten the dipstick tube base nut. See corresponding Service Manual. Referene Procedure 007-011 in Section 7.



Configuration B

Remove

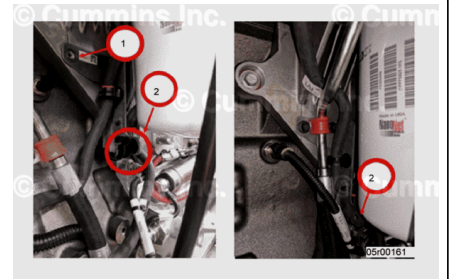
Place an appropriate drip pan or absorbent mat directly under the low-pressure fuel supply tube plumbing to contain any fuel that may discharge during the repair process.

Clean area around the fuel pump gear pump and fuel filter supply tube connections.

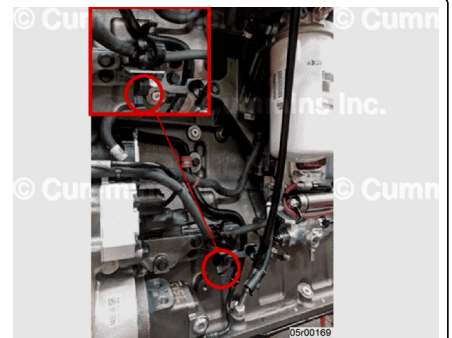
- Disconnect and discard fuel supply tube part number 5440474; plumbing fuel from the low-pressure pump to the fuel filter inlet.
- Remove Quick Disconnect fittings associated with detached fuel supply tube.



- Remove 10mm bolt (1) at filter head upper bracket boss and inspect for re-use. Replace bolt if visual rounding or thread damage is evident. Remove and discard, C-clamp located at the bottommost filter head bracket location (2).

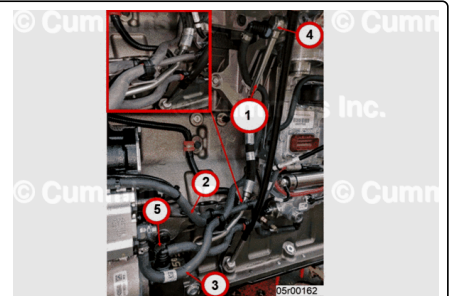


- Remove and discard, C-clamp located at the bottom of fuel supply tube mounting bracket.



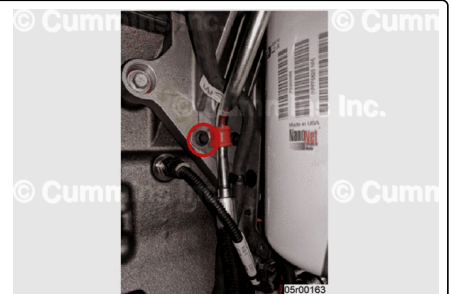
Install

- Route steel fuel supply tube (1) part number 5558764 between fuel pump drain line (2) and filter to fuel pump (3) plumbing.
- Install banjo screw part number 3924725 with 2x sealing washer part number 3963988 hand tight, into fuel supply tube banjo(s):
 - At fuel filter inlet (4).
 - At fuel pump gear pump outlet (5).



- Install P-clip part number 3685612 onto steel fuel supply tube.
- Mount fuel supply tube and P-clip assembly onto fuel filter bracket using 10mm bolt.

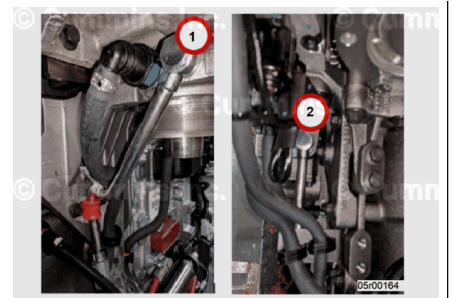
Torque Value: 23 n•m [17 ft-lb]



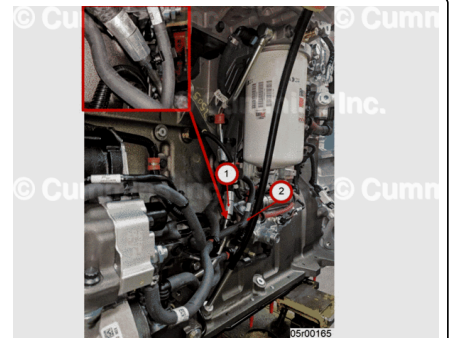
- Torque banjo screws.
 - At fuel filter inlet (1).

- At fuel pump gear pump outlet (2).

Torque Value: 34 n•m [25 ft-lb]



- Install tube clamp; swivel clip part number 3975497, onto steel fuel supply (1) and fuel pump drain (2), at predetermined plumbing location. Ensure there is no contact between fuel supply tubes or engine oil dip stick tube.
- If engine oil dip stick tube is contacting the steel supply tube, clip and reposition looped zip tie adjusting the dipstick tube to provide adequate clearance.

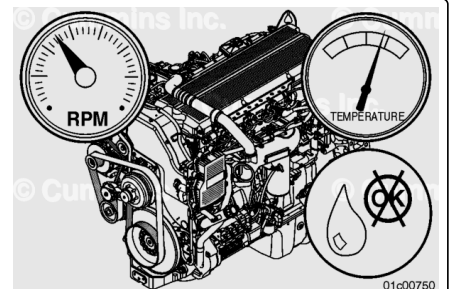


Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. See original equipment manufacturer (OEM) service information.
- Start the engine and check for proper operation. Verify no leaks are present.



Document History

Date	Details
2018-11-5	Module Created
2019-1-14	Drive Clarification to repair locations
2019-1-21	added "Install tube clamp; swivel clip part number 3975497, onto steel fuel supply (1) and fuel pump drain (2), at predetermined plumbing location. Ensure there is no contact between fuel supply tubes or engine oil dip stick tube. If engine oil dip stick tube is contacting the steel supply tube, clip and reposition looped zip tie adjusting the dipstick tube to provide adequate clearance. "
2019-1-31	Image Update - DipStick Tube

